

Exception- Practice- CO3

1.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user "K. Dhilip" with a dropdown menu. The sub-header is "JAVA PROGRAMMING".

On the left, a sidebar shows a list of questions. Question 03, "Finding Exception Using Exception Propagation", is selected and marked as "5/5 answered".

The main area is divided into three sections:

- Your Code:** A code editor showing a Java program:

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int[] a = new int[5];
6         a[0] = 10;
7         a[1] = 20;
8         a[2] = 30;
9         a[3] = 40;
10        a[4] = 50;
11        int index = sc.nextInt();
12        try {
13            System.out.println(a[index]);
14        } catch (ArrayIndexOutOfBoundsException e) {
15            System.out.println("Exception: Invalid array index.");
16        }
17    }
18 }
```
- Input:** A text box containing the value "6".
- Output:** A text box showing the exception message: "Exception: Invalid array index.".
- Visible Test Cases:** Three test cases are listed, all marked as passed with green checkmarks.
- Score:** 10.00 / 10 marks.

A green "Submitted" button is located at the bottom left of the main area.

2.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user "K. Dhilip" with a dropdown menu. The sub-header is "JAVA PROGRAMMING".

On the left, a sidebar shows a list of questions. Question 04, "Finding Exception Using Exception Propagation", is selected and marked as "5/5 answered".

The main area is divided into three sections:

- Your Code:** A code editor showing a Java program:

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int marks = sc.nextInt();
6         try {
7             if (marks < 0 || marks > 100) {
8                 throw new IllegalArgumentException();
9             }
10            System.out.println("Valid marks: " + marks);
11        } catch (IllegalArgumentException e) {
12            System.out.println("Exception: Invalid marks. Marks should be between 0 and 100.");
13        }
14    }
15 }
```
- Input:** A text box containing the value "120".
- Output:** A text box showing the exception message: "Exception: Invalid marks. Marks should be between 0 and 100.".
- Visible Test Cases:** Three test cases are listed, all marked as passed with green checkmarks.
- Score:** 10.00 / 10 marks.

A green "Submitted" button is located at the bottom left of the main area.

3.



SIMATS Online Programming Lab

JAVA PROGRAMMING


K. Didiy
10/10/2023

Q4

Invalid Username

10 marks

Q5

Handling Transaction using Exception Propagation

10 marks

5/5 answered

Your Code

```

1 import java.util.*;
2 public class Main {
3
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6         String password=sc.nextLine();
7         try{
8             if(password.length()<8){
9                 throw new WeakPasswordException(
10                     "Password must contain at least 8 characters."
11                 );
12             }
13             System.out.println("Password is strong.");
14             throw (WeakPasswordException e) {
15                 System.out.println("Exception: " + e.getMessage());
16             }
17         }
18     }
19 }
20 class WeakPasswordException extends Exception {
21     WeakPasswordException(String message) {
22         super(message);
23     }
24 }

```

Submitted

Input

Output

```
Exception: Password must contain at least 8 characters.
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

10.00 / 10 marks

4.



SIMATS Online Programming Lab

JAVA PROGRAMMING


K. Didiy
10/10/2023

Q4

Invalid Username

10 marks

Q5

Handling Transaction using Exception Propagation

10 marks

5/5 answered

Your Code

```

1 import java.util.*;
2 public class Main {
3
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6         String email=sc.nextLine();
7         try{
8             if(email.contains("@")){
9                 throw new InvalidEmailException("Invalid email address");
10             }
11             System.out.println("Valid email address.");
12             throw (InvalidEmailException e) {
13                 System.out.println("Exception: " + e.getMessage());
14             }
15         }
16     }
17 }
18 class InvalidEmailException extends Exception {
19     InvalidEmailException(String message) {
20         super(message);
21     }
22 }

```

Submitted

Input

Output

```
Exception: Invalid email address.
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

10.00 / 10 marks

5.



SIMATS Online Programming Lab
JAVA PROGRAMMING

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2.7 Withdrawal

Your Code

```
1 // imports java.util.*
2
3
4
5 // public class Main {
6
7     static int balance;
8
9     static void withdraw(int amount) throws InsufficientBalanceException {
10         validateBalance(amount);
11     }
12
13     static void validateBalance(int amount) throws InsufficientBalanceException {
14         if (amount > balance) {
15             throw new InsufficientBalanceException();
16             "Withdrawal amount exceeds available balance."
17         }
18     }
19 }
20
21 static void processTransaction(int amount) throws InsufficientBalanceException {
22     withdraw(amount);
23 }
24
25 public static void main(String[] args) {
26
27     Scanner sc = new Scanner(System.in);
28 }
```

Input

Balance:
5000
Withdrawal:
7000

Output

Exception propagated to main().
InsufficientBalanceException
Withdrawal amount exceeds available
balance.

Visible Test Cases

Test Case 1 ✓
Test Case 2 ✓
Test Case 3 ✓

10.00 / 30 marks